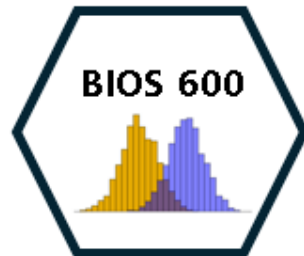


Lecture 8: The central limit theorem

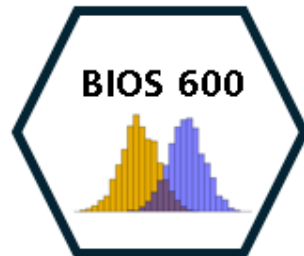
BIOS 600 - Fall 2025

UNC - Chapel Hill



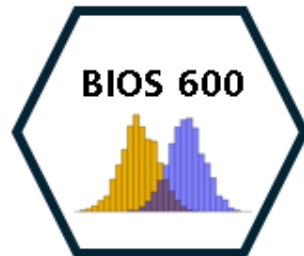
Announcements

- ▶ Tuesday 9/16 Lab Session is just TA office hours. Swing by if you need help on HW 3.
- ▶ HW 3 due Thursday 9/18 at 11:59pm.
- ▶ Thursday 9/18 class will be a review session.
- ▶ Tuesday 9/23: Exam 01 in-class. All you need is something to write with.



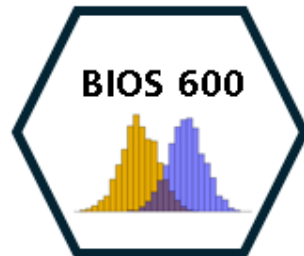
Overview

- ▶ Central Limit Theorem
- ▶ Application Exercise
- ▶ Why is the Central Limit Theorem important?



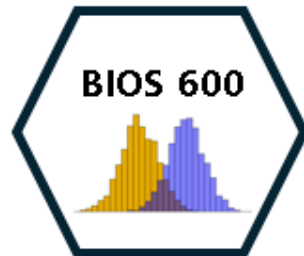
Reading

- ▶ Pagano and Gavreau: Sections 8.1-8.4
- ▶ [OpenIntro Statistics](#): Section 5.1



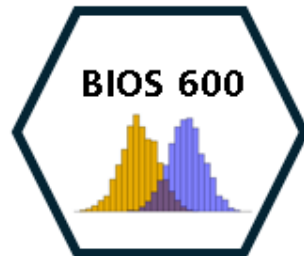
A roadmap ahead

- ▶ In just a few more lectures, we'll have learned enough tools to perform a pretty wide range of analyses and answer associated questions.
- ▶ In the latter half of this course, we'll emphasize specific scientific questions of interest, statistical methods for testing such questions, and translating those results to real-world conclusions.



A roadmap for today

- ▶ Today's topic is fundamental to making sound statistical inferences and conclusions.
- ▶ The basic idea is called the **central limit theorem**, which states that for **any** distribution with a well-defined mean and variance, the distribution of the means computed from samples of size n will be approximately Gaussian.
normally distributed



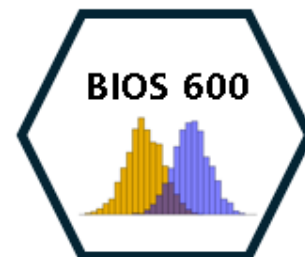
Recall: What is statistical inference?

- ▶ Statistical inference is the act of generalizing from a **sample** in order to make conclusions regarding a **population** while quantifying the degree of certainty we have.
- ▶ We are interested in population **parameters**, which we do not observe.
- ▶ Instead, we must calculate **statistics** from our sample in order to learn about the parameters.

"truth"

population mean μ

↓ example: Sample mean $\bar{X}$

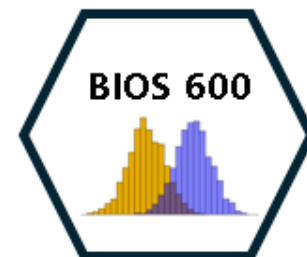
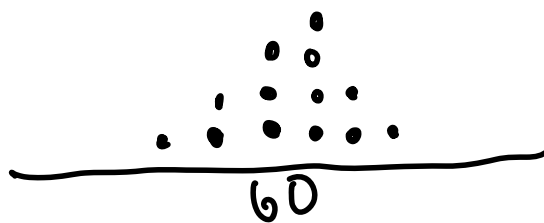


The sampling distribution of the mean

Suppose we're interested in the resting heart rate of students in BIOS 600, and are able to do the following:

1. Take a random sample of size n from this population
 - ▶ Calculate the mean resting heart rate *in this sample*, $\bar{x}_1$
2. Put the sample back, take a second random sample of size n
 - ▶ Calculate the mean resting heart rate from this new sample, $\bar{x}_2$
3. Put the sample back, take a third random sample of size n
 - ▶ Calculate the mean resting heart rate from this sample, too...

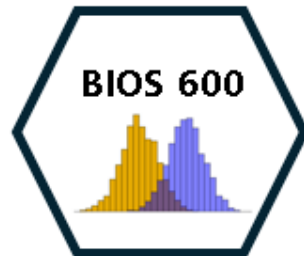
... and so on



The sampling distribution of the mean

After repeating this many times, we have a dataset that has the sample averages from the population: $\{\bar{x}_1, \bar{x}_2, \dots\}$

- Can we say anything about the distribution of these sample means?

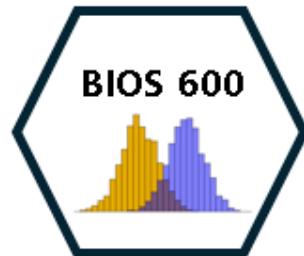


The central limit theorem

The **central limit theorem** states that for a population with mean μ and standard deviation σ , these three properties hold for the distribution of sample averages $\bar{X}$:

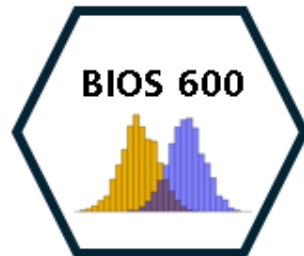
1. The mean of the sampling distribution is identical to the population mean μ
2. The standard deviation of the distribution of the sample averages is $\sigma/\sqrt{n}$, or the **standard error** (SE) of the mean
3. For n large enough (in the limit, as $n \rightarrow \infty$), the shape of the sampling distribution of means is approximately normal

*n=20
in HP
example*



What is this really saying?

- ▶ The central limit theorem tells us that **sample averages** are normally distributed, if we have enough data.
- ▶ This is true **even if** our original variables are not normally distributed.

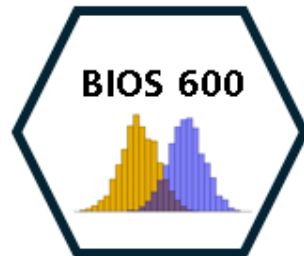


Application Exercise

Application Exercise

For today's participation, go to Canvas and find instructions and template for AE 02.

AE 02 is due on Canvas on **Friday 9/19 at 11:59pm.**



Why the CLT matters

- ▶ **Gives us Normality (for free!)**

- ▶ Even if the population distribution is not normal, the sample mean is approximately normal when sample size is large enough.

- ▶ **Foundation for Inference**

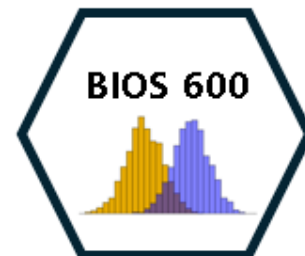
- ▶ Lets us use normal-based methods for **confidence intervals** and **hypothesis tests**.
 - ▶ Justifies the use of t-tests, z-tests, ANOVAs, regression, etc.

- ▶ **Works Across Many Situations**

- ▶ Whether data are binary (yes/no), counts, or skewed — sample averages will still tend toward a bell curve.

- ▶ **Practical Takeaway**

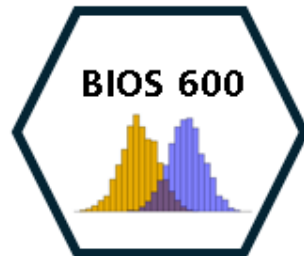
- ▶ We can make powerful statistical conclusions **without needing to know the exact population distribution**.



Example: IQ tests

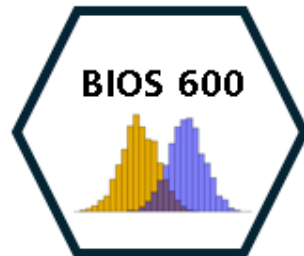
- ▶ IQ tests are designed to have a probability distribution with $\mu = 100$ and $\sigma = 15$.
- ▶ Suppose we draw samples of size $n = 20$ from this population.
- ▶ From the central limit theorem, the distribution of the sample averages will be approximately normal with mean 100 and standard deviation $15/\sqrt{20}$

$$\underbrace{15/\sqrt{20}}_{\parallel} \sigma/\sqrt{n}$$



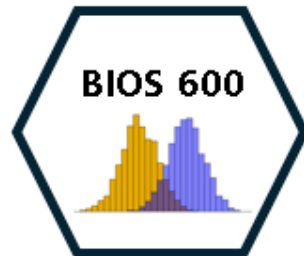
IQ tests

- ▶ If the population distribution is normal to begin with, then the distribution of the sample averages will also be exactly normal.
- ▶ If the population distribution is not normal, then the rule of thumb is that we need at least $n = 30$ for the central limit theorem to kick in for approximate normality.



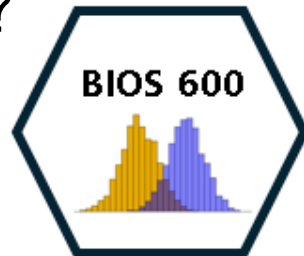
Example

- ▶ Suppose I give a random sample of $n = 30$ BIOS 600 students an IQ test*, and the sample average score is 120.
- ▶ Does this mean that BIOS 600 students are smarter than average?
- ▶ I know there are many problems with IQ and IQ testing...bear with me here!

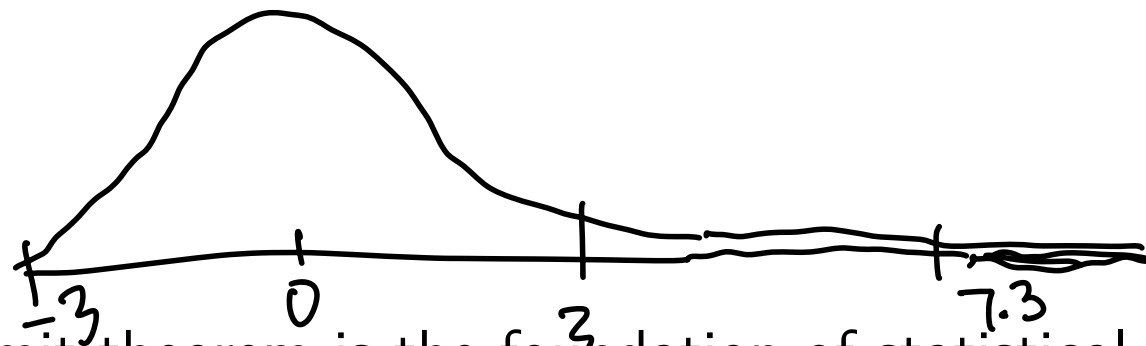


Example

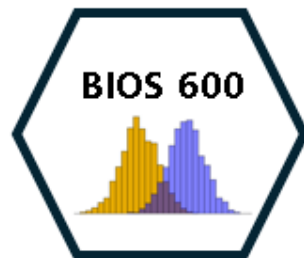
- ▶ The central limit theorem tells us that the distribution of means of samples of size 30 from this population is also normal, with mean $\mu = 100$ and $SE = \sigma/\sqrt{n} = 15\sqrt{30} \approx 2.7$.
- ▶ $Z = \frac{\bar{X} - \mu}{SE}$ is a standard normal random variable, and here $Z \approx 7.3$. *← standardizing to produce z-score*
- ▶ The probability of a z-score greater than this is extremely small.
- ▶ What does this mean? (p-values coming soon...)
- ▶ What are the upper and lower limits that enclose 95% of the means for samples of size n drawn from the population? (Confidence intervals coming soon...)



Recap



- ▶ The central limit theorem is the foundation of statistical inference.
- ▶ The Central Limit Theorem tells us that when you take the average of a large enough sample, that average will follow a normal distribution, even if the original data don't.
- ▶ This is important because many statistical methods, like confidence intervals and hypothesis tests, rely on normal distributions.
- ▶ Thanks to the CLT, we can apply these methods to real-world data, even if the data itself isn't normally distributed, as long as the sample size is large enough.



Next class

- ▶ Review for Exam 01

